TEST PLAN

OrangeHRM Open Source Testing Project

Document Information

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Prepared By	Ali Muhammed
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1. Introduction

This Test Plan defines the testing strategy, objectives, scope, resources, schedule, and deliverables for the OrangeHRM Open Source application.

The purpose of this document is to ensure that the application functions correctly, meets business requirements, and provides a reliable user experience through comprehensive testing activities.

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2. Project Overview

OrangeHRM is a Human Resource Management System (HRMS) used to manage employee information, recruitment processes, leave requests, user administration, and employee records.

The project will be tested using multiple testing approaches including Manual Testing, API Testing, Database Testing, and Automation Testing.

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3. Testing Objectives

The main objectives of this testing project are:

- Verify system functionality against requirements.
- Identify and report software defects.
- Validate data integrity within the database.
- Verify API functionality and responses.
- Ensure critical business processes work correctly.
- Improve application quality and reliability.
- Build a complete QA portfolio project following industry standards.

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4. Scope of Testing

In Scope

Authentication Module

- Login
- Logout
- Forgot Password
- Password Validation

Admin Module

- User Management
- Add User
- Edit User
- Delete User
- Search User
- User Roles

PIM Module

- Add Employee
- Edit Employee

- Delete Employee
- Employee Search
- Employee Details

Leave Module

- Apply Leave
- Approve Leave
- Reject Leave
- Cancel Leave
- Leave Reports

Recruitment Module

- Add Candidate
- Edit Candidate
- Candidate Search
- Vacancy Management

Dashboard Module

- Dashboard Widgets
- Quick Launch Features

API Testing

- Authentication APIs
- Employee APIs
- Leave APIs
- Recruitment APIs

Database Testing

- Employee Data Validation
- User Data Validation
- Leave Data Validation

- CRUD Operations Verification

Out of Scope

- Mobile Application Testing
- Performance Testing
- Load Testing
- Penetration Testing
- Third-Party Integrations

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5. Testing Types

Functional Testing

Verify that all system functionalities work according to business requirements.

Activities:

- Positive Testing
- Negative Testing
- Boundary Value Testing
- Exploratory Testing

UI Testing

Verify:

- Labels
- Buttons
- Forms
- Layout Consistency
- Navigation
- Error Messages

Smoke Testing

Verify critical functionalities after deployment:

- Login
- Add Employee
- Employee Search
- Apply Leave
- Logout

Regression Testing

Verify that existing functionality remains unaffected after updates or bug fixes.

API Testing

Tool: Postman

Validation Areas:

- Status Codes
- Response Body
- Response Headers
- Authorization
- Response Time
- Error Handling

Methods Covered:

- GET
- POST
- PUT
- DELETE

Database Testing

Tool: MySQL

Validation Areas:

- Data Accuracy
- Data Consistency
- CRUD Operations
- Foreign Key Relationships
- Duplicate Records
- Data Integrity

Sample Checks:

- Employee creation updates database correctly.
- Employee deletion removes records correctly.
- Leave requests are stored accurately.

Automation Testing

Tools:

- Selenium WebDriver
- Java
- TestNG
- Maven

Framework Design:

- Page Object Model (POM)
- Data Driven Testing
- Reusable Utilities
- Extent Reports

- Screenshot Capture

Automated Scenarios:

- Login
- Logout
- Add Employee
- Employee Search
- Apply Leave

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6. Test Environment

Hardware

- Personal Computer
- Minimum 8 GB RAM

Operating System

- Windows 10
- Windows 11

Browsers

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Tools

- Jira
- Postman
- MySQL Workbench
- Selenium WebDriver
- TestNG
- Maven

- GitHub

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7. Test Deliverables

The following artifacts will be produced during the project:

Documentation

- Test Plan
- Test Scenarios
- Test Cases
- RTM
- Test Summary Report

Defect Management

- Bug Reports
- Defect Tracking Log

API Testing

- Postman Collection
- Environment Files
- Newman Reports

Database Testing

- SQL Queries
- Validation Reports

Automation Testing

- Selenium Framework
- Test Execution Reports
- Screenshots

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8. Entry Criteria

Testing activities may begin when:

- Application is deployed successfully.
- Test environment is available.
- Test data is prepared.
- Required modules are accessible.

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9. Exit Criteria

Testing activities will be completed when:

- All planned test cases have been executed.
- Critical defects are resolved.
- Smoke Test passes successfully.
- Test Summary Report is completed.
- Project deliverables are finalized.

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10. Risk Assessment

Risk	Impact	Mitigation
Environment instability	Testing delays	Retest after environment recovery
Requirement changes	Test case updates	Review documentation regularly
Insufficient test data	Limited coverage	Create dedicated test datasets
Browser compatibility issues	User impact	Execute cross-browser testing

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11. Roles and Responsibilities

Role	Responsibility
QA Engineer	Planning, Execution, Reporting

Role	Responsibility
Application Under Test	OrangeHRM Open Source
Project Repository	GitHub

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12. Success Metrics

The project will be considered successful if:

- Requirement Coverage ≥ 95%
- Test Case Execution Rate ≥ 95%
- Critical Defects = 0
- High Severity Defects Closed
- Core Business Flows Successfully Automated

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13. Estimated Project Workflow

Phase 1:

- Test Plan
- Requirement Analysis

Phase 2:

- Test Scenarios Creation

Phase 3:

- Test Cases Design

Phase 4:

- Test Execution

Phase 5:

- Bug Reporting

Phase 6:

- RTM Preparation

Phase 7:

- API Testing

Phase 8:

- Database Testing

Phase 9:

- Automation Testing

Phase 10:

- Test Summary Report

Approval

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